COS30049 – Computing Technology Innovation Project

Week 3 - Monitoring, Controlling, and Closure in Project Management & Usability of a Web Project

(Lecture – 02)

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Usability of a Web Project



Image by Ben Kolde on Unsplash

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Definition of Usability



Definition:

Usability is the measure of how easy and efficient a system, tool, or device is to use. It refers to the ability of users to effectively learn and utilize the design to achieve their goals with satisfaction.

Key Aspects of Usability:

- Learnability: How quickly can a new user learn to use the interface or system?
- Efficiency: Once learned, how quickly can users accomplish their tasks?
- Memorability: How easily can users remember how to use the system after not using it for a while?
- Error Handling: How many errors do users make, and how easily can they recover from them?
- Satisfaction: How pleasant is it to use the design?

Importance in Web Projects:

- Enhances user satisfaction.
- Increases efficiency and effectiveness
- Reduces errors and improves overall user experience.

Why UX research is important



Image Source:[30 Funniest UX Memes," PlaybookUX]

Widely Recognized Usability Principles

Usability principles guide the design of user-friendly interfaces. Several widely recognized principles include:

- Ben Shneiderman's Eight Golden Rules of Interface Design.
- Nielsen's Heuristics for User Interface Design
- Gerhardt-Powals' Cognitive Engineering Principles
- ISO 9241-11: Ergonomic requirements for office work with visual display terminals (VDTs)



Photo by David Travis on Unsplash

"8 Golden Rules of Interface Design"

Shneiderman – from text

- In general, sets of principles are in close agreement
  - E.g., Shneiderman, Nielsen, and Togazzini
- **Schneiderman's 8 rules (principles):**
  1. Strive for consistency
  2. Cater to universal usability
  3. Offer informative feedback
  4. Design dialogs to yield closure
  5. Prevent errors
  6. Permit easy reversal of actions
  7. Support internal locus of control
  8. Reduce short term memory

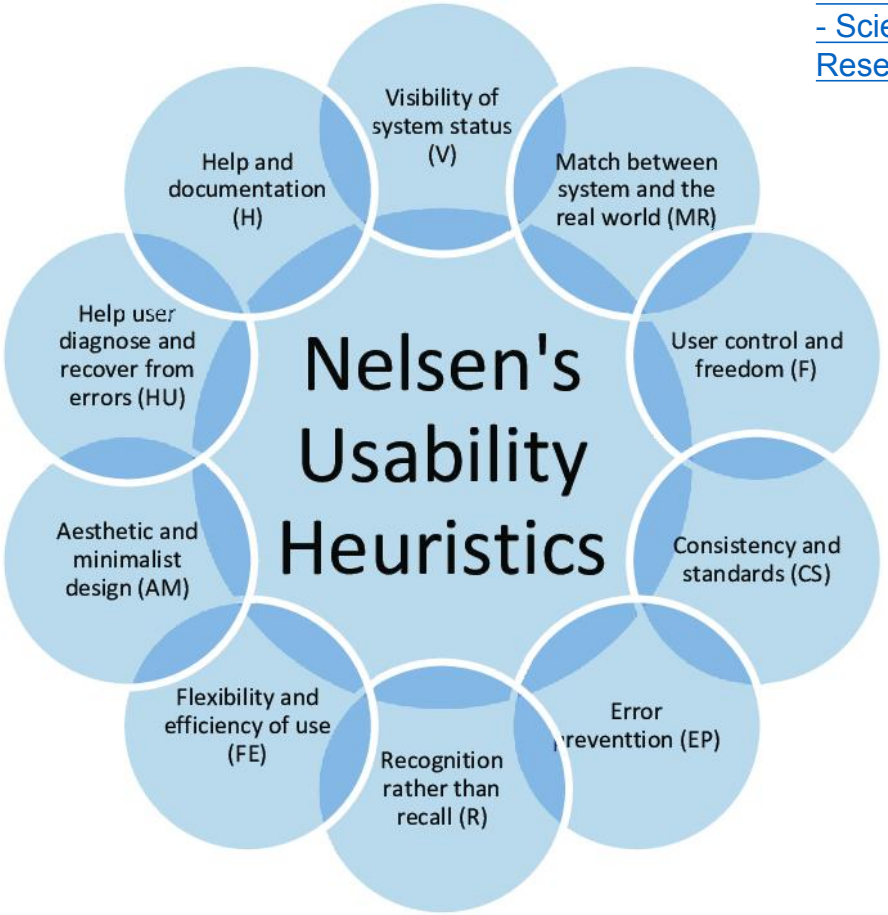
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Principles - Nielsen's Usability Heuristics (A1)

What's this?

- Jakob Nielsen's heuristics are widely used in the usability field to evaluate and design user interfaces. They provide a robust framework for understanding and improving usability.

Image Source:[Usability evaluation of e-government websites: A case study from Taiwan - Scientific Figure on ResearchGate]



Ten Usability Heuristics

- Visibility of system status
- Match between system and the real world
- User control and freedom
- Consistency and standards
- Error prevention
- Recognition rather than recall
- Flexibility and efficiency of use
- Aesthetic and minimalist design
- Help users recognize, diagnose, and recover from errors
- Help and documentation

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Principles - Shneiderman's Eight Golden Rules

What's this?

Shneiderman's Eight Golden Rules of Interface Design are a set of principles aimed at creating user-friendly, consistent, and efficient interfaces that enhance the overall user experience by minimizing errors reducing cognitive load, and encouraging intuitive interaction.



Eight Golden Rules:

- Strive for Consistency
- Enable Frequent Users to Use Shortcuts
- Offer Informative Feedback
- Design Dialogs to Yield Closure
- Offer Simple Error Handling
- Permit Easy Reversal of Actions
- Support Internal Locus of Control
- Reduce Short-Term Memory Load

Image Source:[Tripti Rajput, "Analyzing Instagram Using Shneiderman's Eight Golden Rules of Interface Design," Medium]

Usability Evaluation Methods

Types:

- Usability Testing: Observing real users as they interact with the product.
- Heuristic Evaluation: Experts review the product based on established usability principles.
- Surveys and Questionnaires: Collecting user feedback.

Heuristic evaluation vs. usability testing

HEURISTIC EVALUATION		USABILITY TESTING
An expert review of the interface against established usability principles	**Definition**	Real users performing tasks to identify usability issues
UI designers and product specialists (evaluators)	**Participants**	End-users, often matching the product's ideal user persona
Qualitative analysis based on pre-determined heuristics	**Methodology**	Empirical testing involving task completion and user observation
Less prone to user bias; relies on expert evaluation	**Bias Consideration**	Risk of data being skewed by existing users' familiarity with the product
Not applicable as it's an expert-driven process	**Remote Feasibility**	Highly feasible, allowing for a broader user reach and faster insights
Primarily used early in the design process for quick expert insights	**Complementary Nature**	Often used in conjunction with heuristic analysis for comprehensive evaluation

Image Source: Maze Guide on Heuristic

Usability Testing Methods

Types:

- Moderated Testing: Involves a facilitator guiding participants through tasks.
- Unmoderated Testing: Participants complete tasks on their own without a facilitator.
- Remote Testing: Conducted over the internet with participants in different locations.
- In-Person Testing: Conducted in a controlled environment with direct observation.

Process:

- Planning:
  - Define objectives and metrics.
  - Select participants.
- Conducting Tests:
  - Create tasks based on real user scenarios.
  - Observe and record user interactions.
- Analysis:
  - Identify usability issues and patterns.
  - Summarize findings and make recommendations.

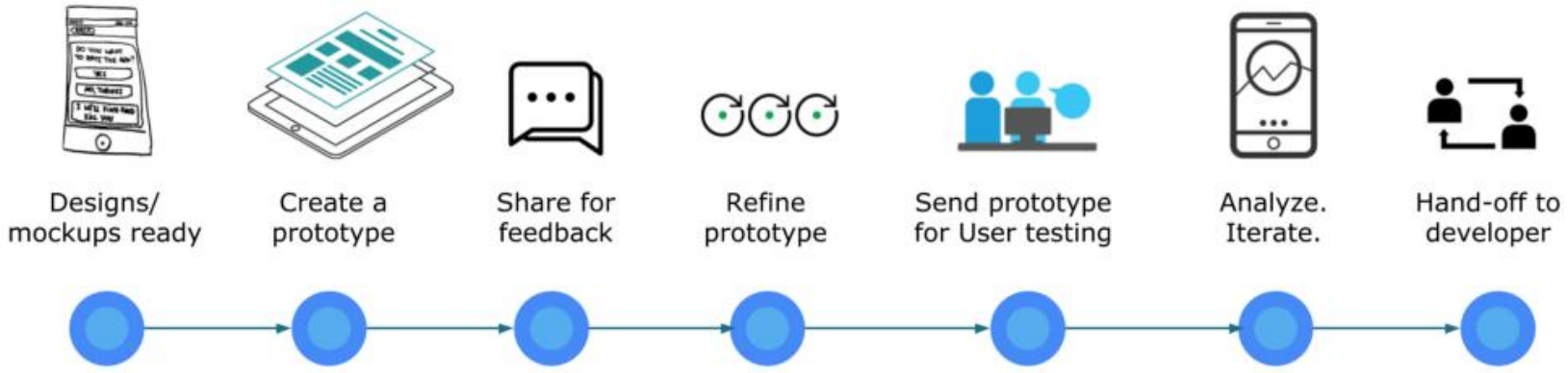


Image Source: a chart that presents the different steps in usability testing can easily be replicated)

Tools for Usability Testing

Popular Usability Testing Tools:

1. User Testing:

- Purpose: Conduct remote usability tests with real users to gather insights on how they interact with your product.
- Features:
  - Real-time video feedback from users.
  - Ability to test on various devices (desktop, mobile, tablet).
  - Instant access to user demographics.
- Use Case: Ideal for testing new website designs or mobile apps to understand user behavior and identify pain points.

2. Optimal Workshop:

- Purpose: Provides tools for card sorting and tree testing to help you understand how users organize and navigate your content.
- Features:
  - Card sorting for information architecture design.
  - Tree testing to evaluate how easily users can find information.
  - Participant recruiting and reporting tools.
- Use Case: Useful for organizing complex information structures, like website menus or product categories.

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Tools for Usability Testing

3. Hotjar:

- Purpose: Offers heatmaps and session recordings to analyze user behavior on your site.
- Features:
  - Heatmaps to visualize where users click, move, and scroll.
  - Session recordings to see exactly how users navigate through your website.
  - Feedback polls and surveys to gather user input.
- Use Case: Best for identifying areas of interest and friction on your site, allowing you to optimize user experience.

4. Crazy Egg:

- Purpose: Provides visual representations of user interactions through click maps and scroll maps.
- Features:
  - Click maps to see where users are clicking on your site.
  - Scroll maps to understand how far down users scroll on a page.
  - A/B testing to compare different versions of a page.
- Use Case: Great for improving the layout and design of landing pages to increase conversion rates.

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User-Centered Design Process

- **Definition**: User-Centered Design (UCD) is an iterative design process that focuses on understanding the users' needs, preferences, and limitations throughout all phases of the design process.
- **Purpose**: The goal is to create products that are both useful and usable by involving users in the design process.

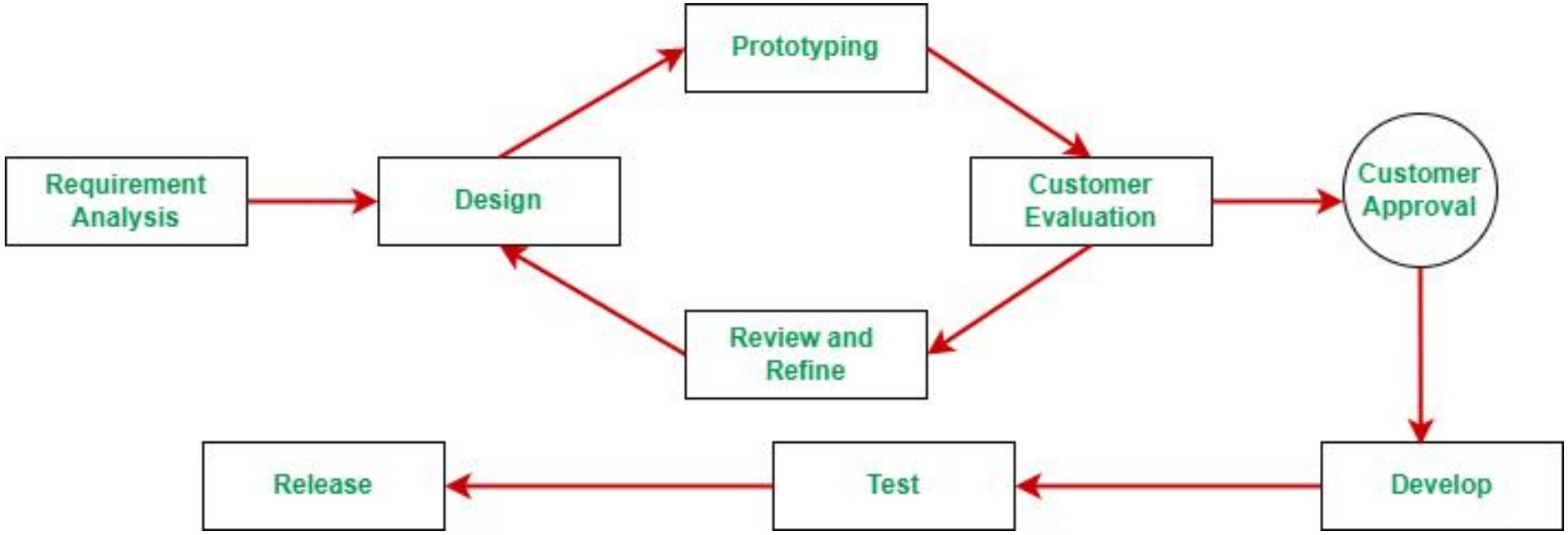


Image Source: prototype testing in software testing on geeksforgeeks.org)

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Apply Usability Concepts with Real Examples



We want user experience to go like this...



But it usually ends up like this...

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Image Source:[30 Funniest UX Memes," PlaybookUX]

Web Usability Examples - Bad

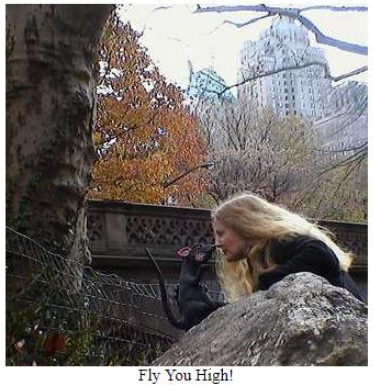
Suzanne Collins

Home Works Biography Interview

Welcome

Hi! Thanks for visiting my website. Here's a picture of me with a rat in Central Park. If you've read my fantasy series, *The Underland Chronicles*, you will have a clue as to why I chose this photo. If not, you may want to click around and find out a little more about my books. Or visit the Scholastic website for cool games and info!

Click here to check out Scholastic's Underland Chronicles website!



Fly You High!



MOVIE TIE-IN EDITION IS NOW AVAILABLE!

#1 USA Today Bestseller

Selected Works

Picture Book

YEAR OF THE JUNGLE
"Important and necessary."
--Kirkus Reviews, STARRED REVIEW

WHEN CHARLIE MCBUTTON LOST POWER
"A clever, humorous story in rhyme."
--School Library Journal

Sci-Fi

MOCKINGJAY
"...every bit as original and thought provoking, as *The Hunger Games*. Wow."
--Los Angeles Times

CATCHING FIRE
"...doesn't disappoint when it segues into the pulse-pounding action readers have come to expect."
--Publishers Weekly, STARRED REVIEW

THE HUNGER GAMES
"...a superb tale of physical adventure, political suspense, and romance."
--Booklist, STARRED REVIEW

Fantasy

GREGOR THE OVERLANDER
"...readers will likely find [the Underland] to be a fantastically engaging place."
--Publishers Weekly, STARRED REVIEW

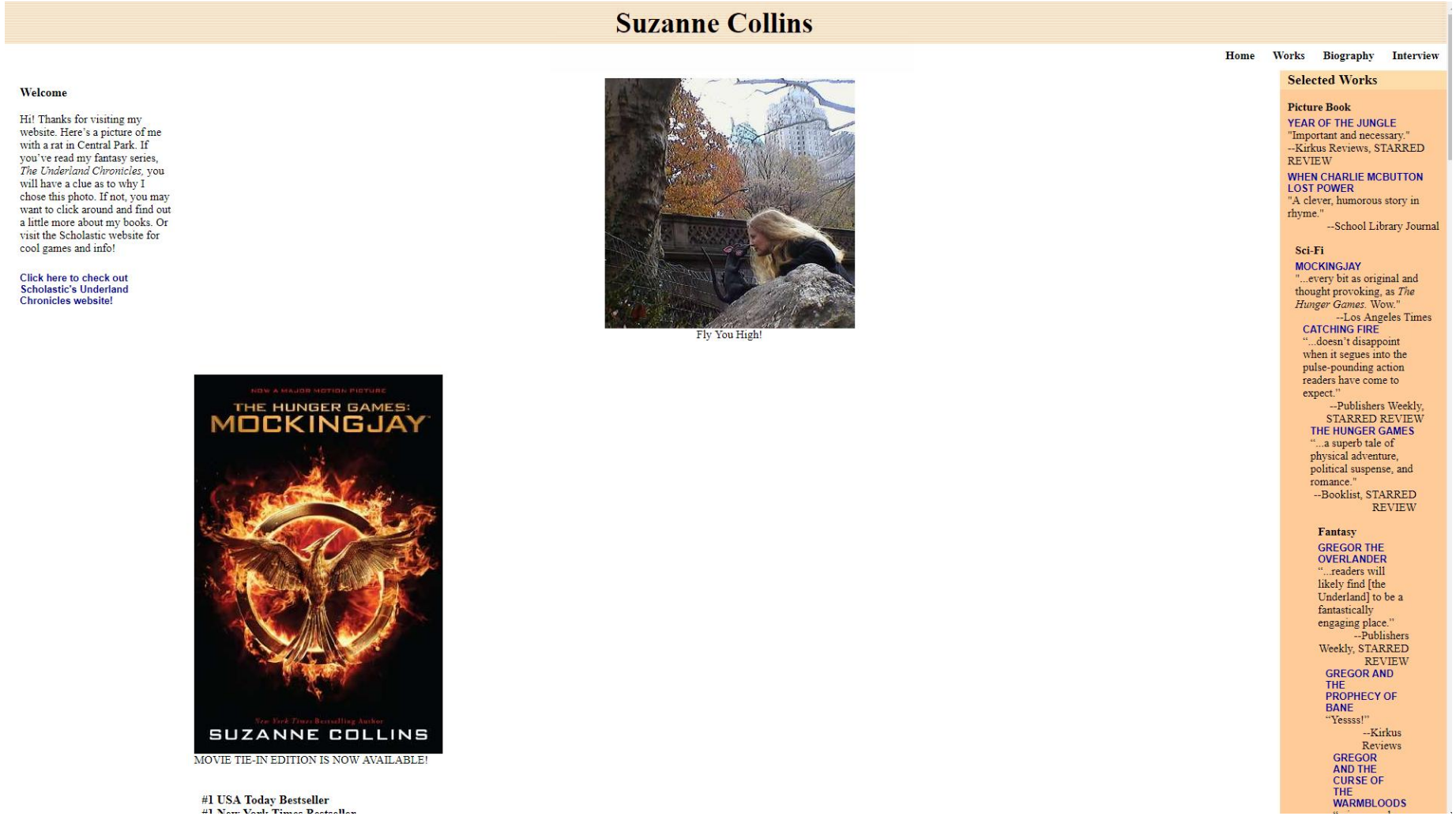
GREGOR AND THE PROPHECY OF BANE
"Yessss!"
--Kirkus Reviews

GREGOR AND THE CURSE OF THE WARMBLOODS

Image Source:["21 Bad Website Examples," Weblium]

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Web Usability Examples - Bad



1. Excessive White Space
2. Poor Visual Hierarchy
3. Text Readability
4. Outdated Design
5. Inconsistent and Cluttered Content

Image Source:["21 Bad Website Examples," Weblium]

Web Usability Examples - Bad



If you want to visit this website: http://arngren.net/

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Web Usability Examples - Bad



1. Poor navigation.
2. Tiny and even confusing typography.
3. The absence of a clear message about the business.
4. Random use of colors……..

If you want to visit this website: http://arngren.net/

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Web Usability Examples - Bad

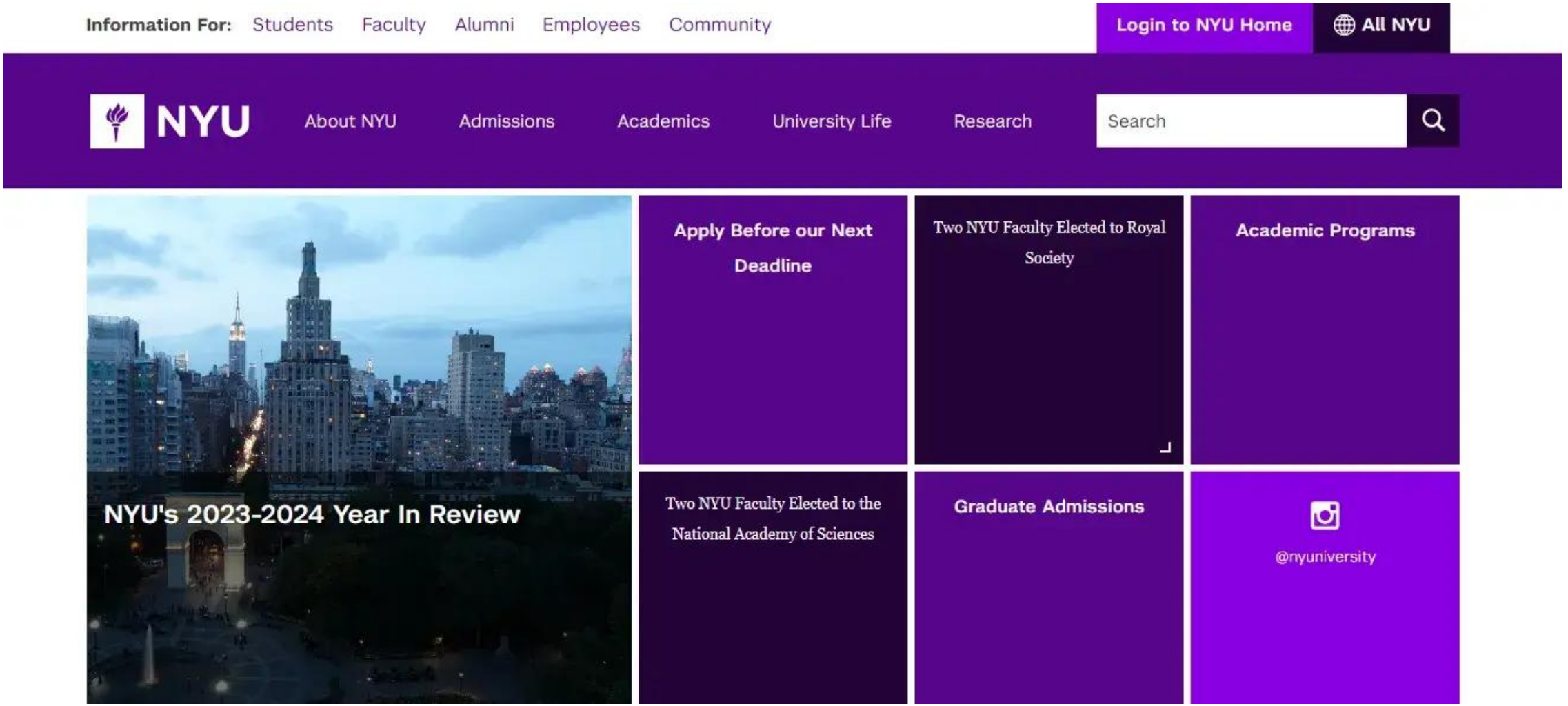
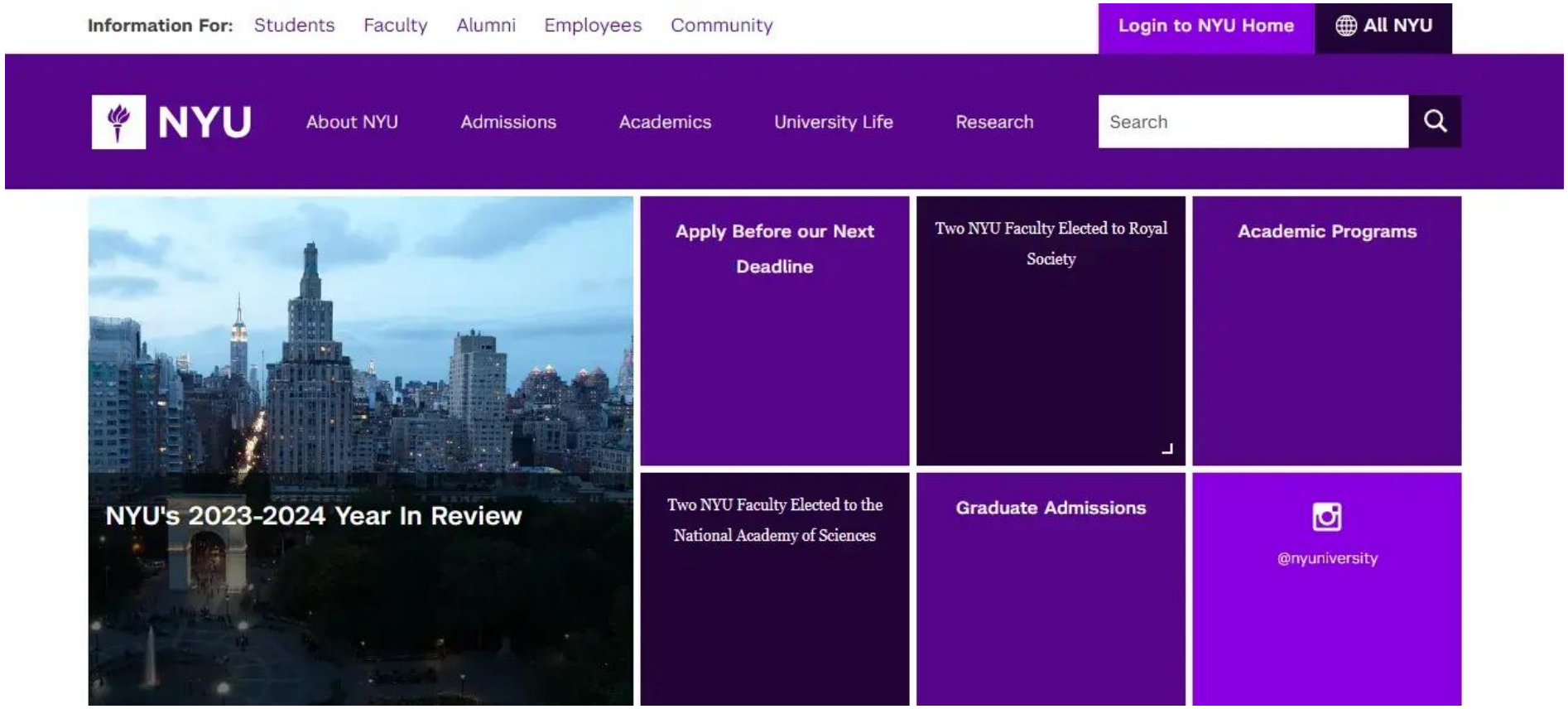


Image Source:["Bad vs. Good Design: What's the Difference?" HubSpot Blog]

Web Usability Examples - Bad



Not That Bad …. But:

- Low contrast on NYU's homepage hinders navigation
- The grid layout gives the impression that the webpage isn't fully loaded
- More images could enhance clarity

Image Source:["Bad vs. Good Design: What's the Difference?" HubSpot Blog]

Nielsen's Usability Heuristics with Examples

1. Visibility of System Status

The system should always keep users informed about what is going on, through appropriate feedback within a reasonable time.

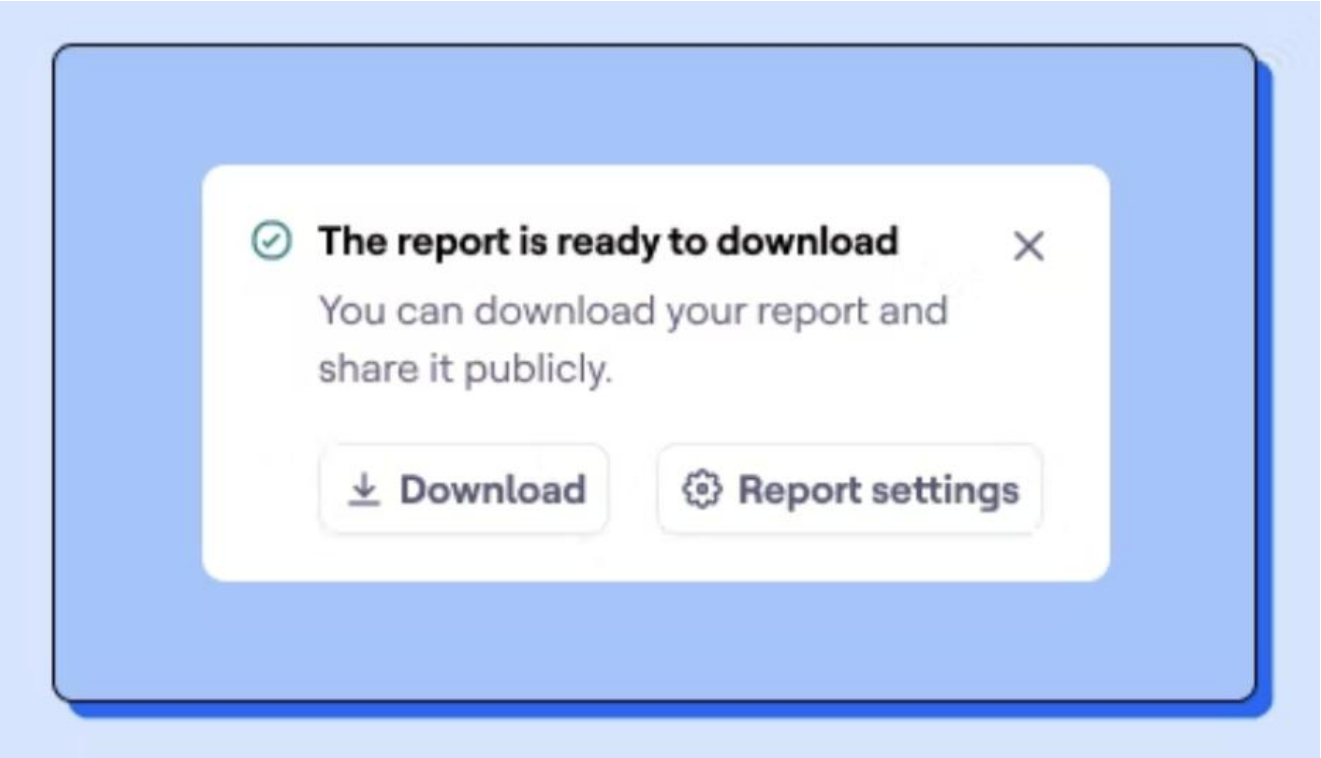


Image Source: Maze Guide on Heuristic Evaluation

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Nielsen's Usability Heuristics with Examples

2. Match Between System and the Real World

The system should speak the users' language, with words, phrases, and concepts familiar to the user, rather than system-oriented terms.

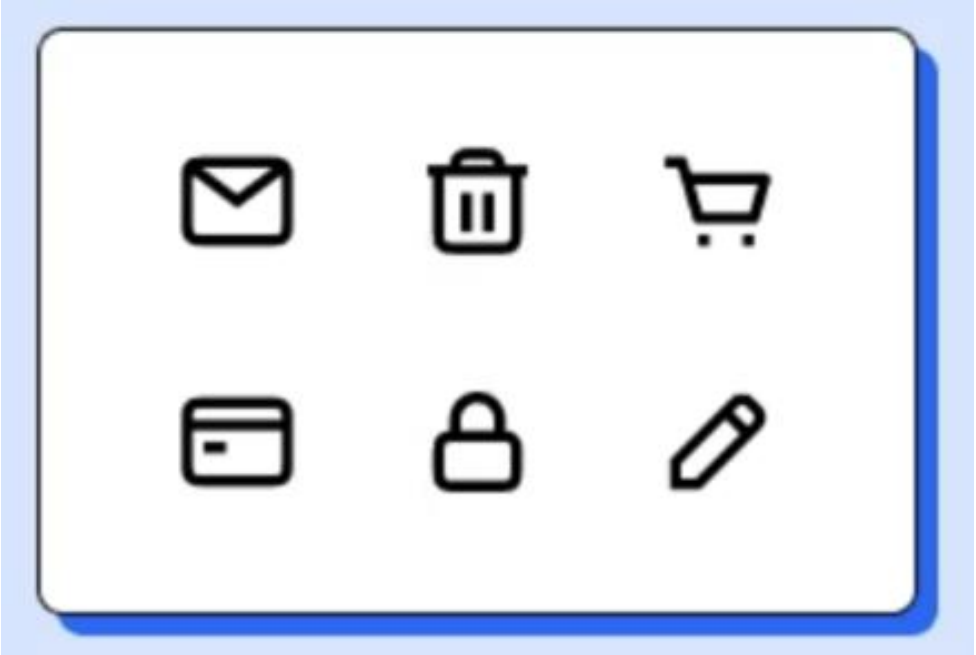


Image Source: Maze Guide on Heuristic Evaluation

Nielsen's Usability Heuristics with Examples

3. User Control and Freedom

Users often choose system functions by mistake and need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue.

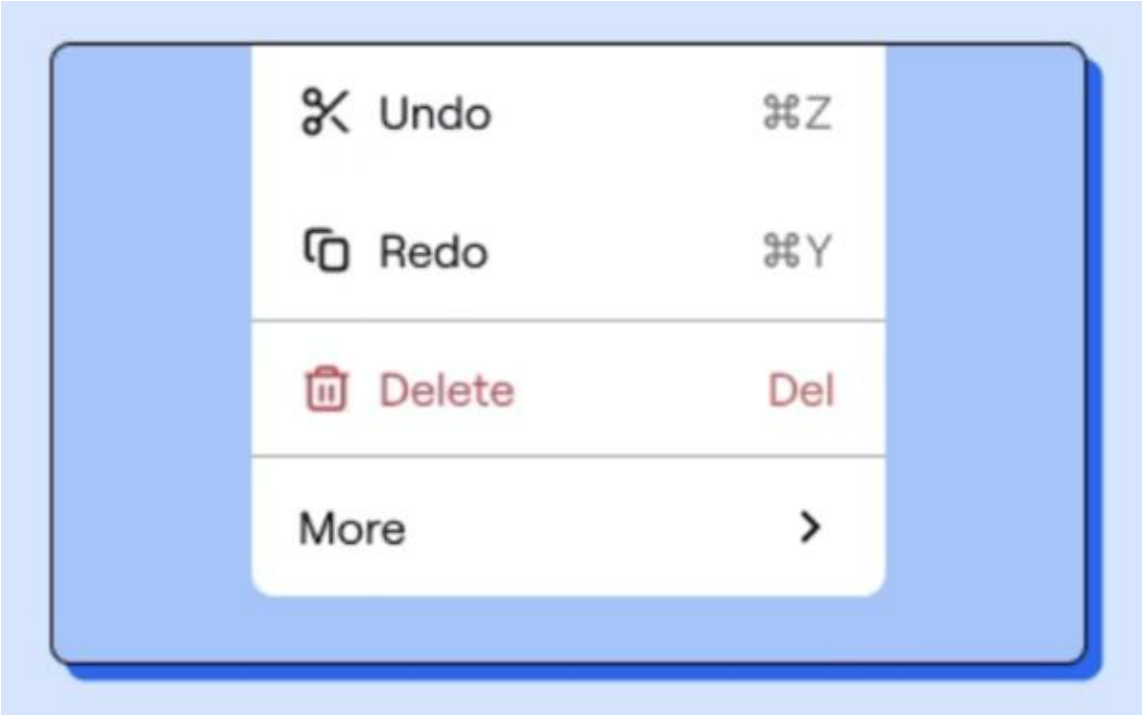


Image Source: Maze Guide on Heuristic Evaluation

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Nielsen's Usability Heuristics with Examples

4. Consistency and Standards

Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.

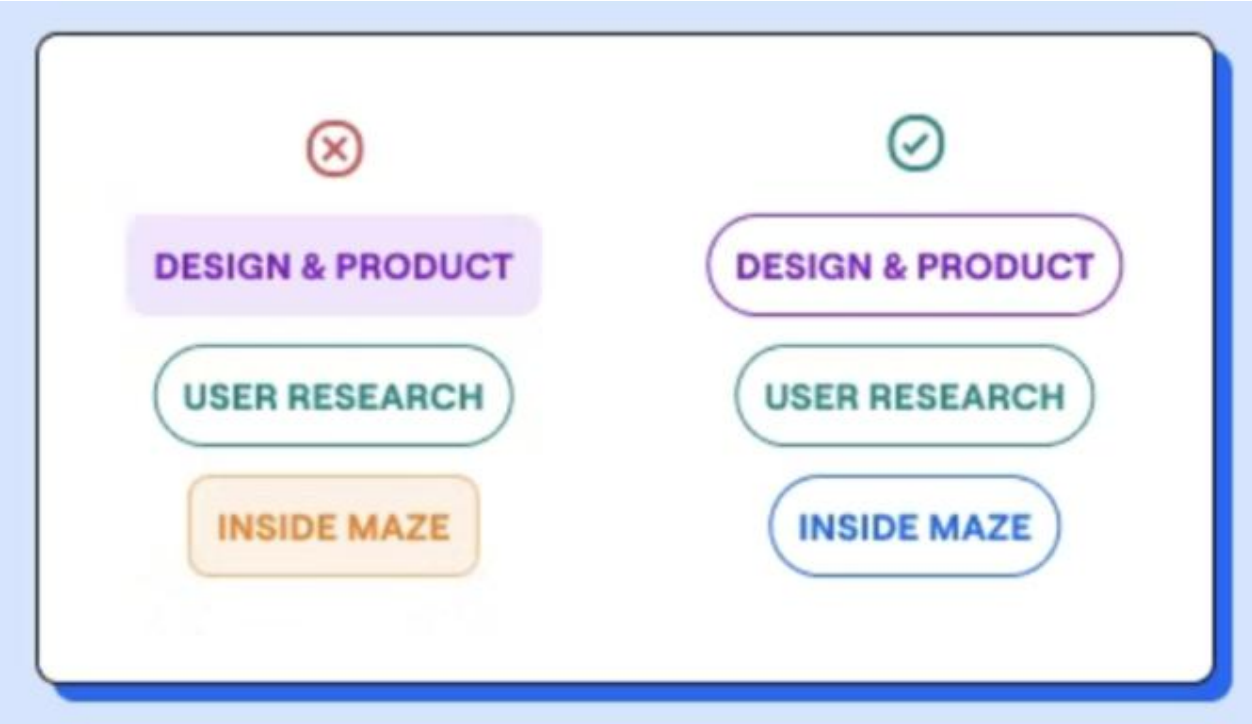


Image Source: Maze Guide on Heuristic Evaluation

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Nielsen's Usability Heuristics with Examples

5. Error Prevention

Even better than good error messages is a careful design that prevents a problem from occurring in the first place.



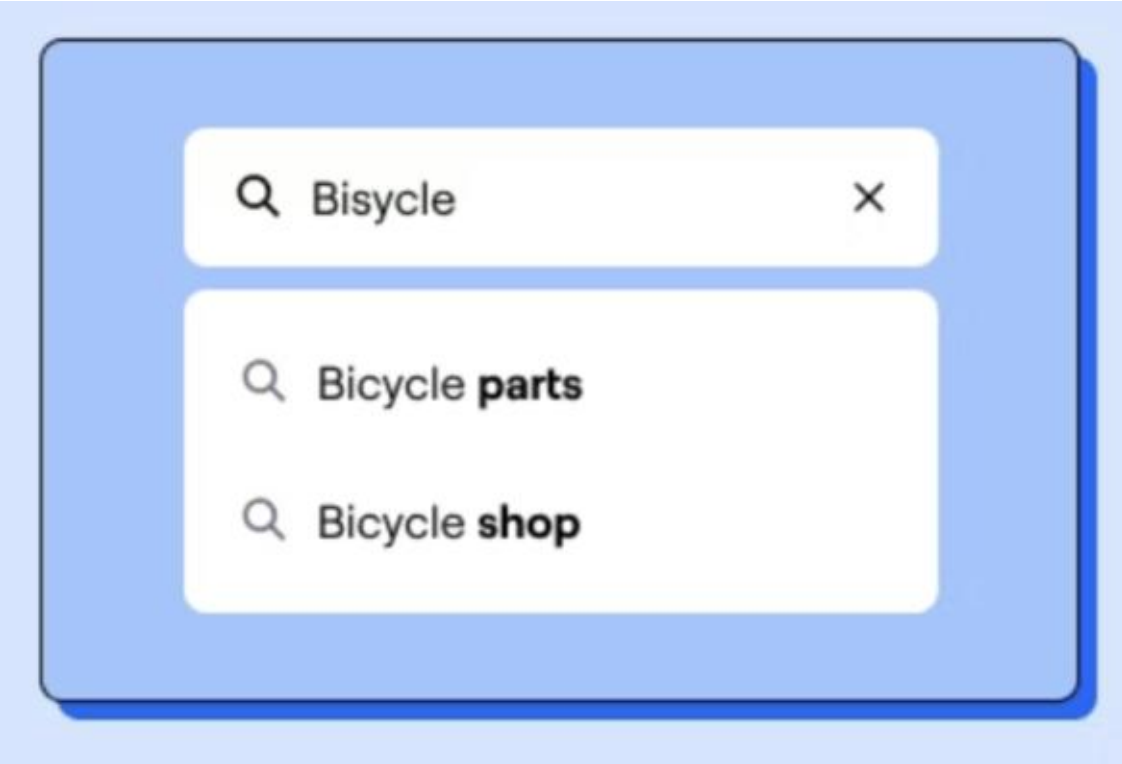


Image Source: [Nielsen's Usability Heuristics for User Interface Design," UX Design]

Image Source: Maze Guide on Heuristic Evaluation

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Nielsen's Usability Heuristics with Examples

6. Recognition Rather Than Recall

Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another.

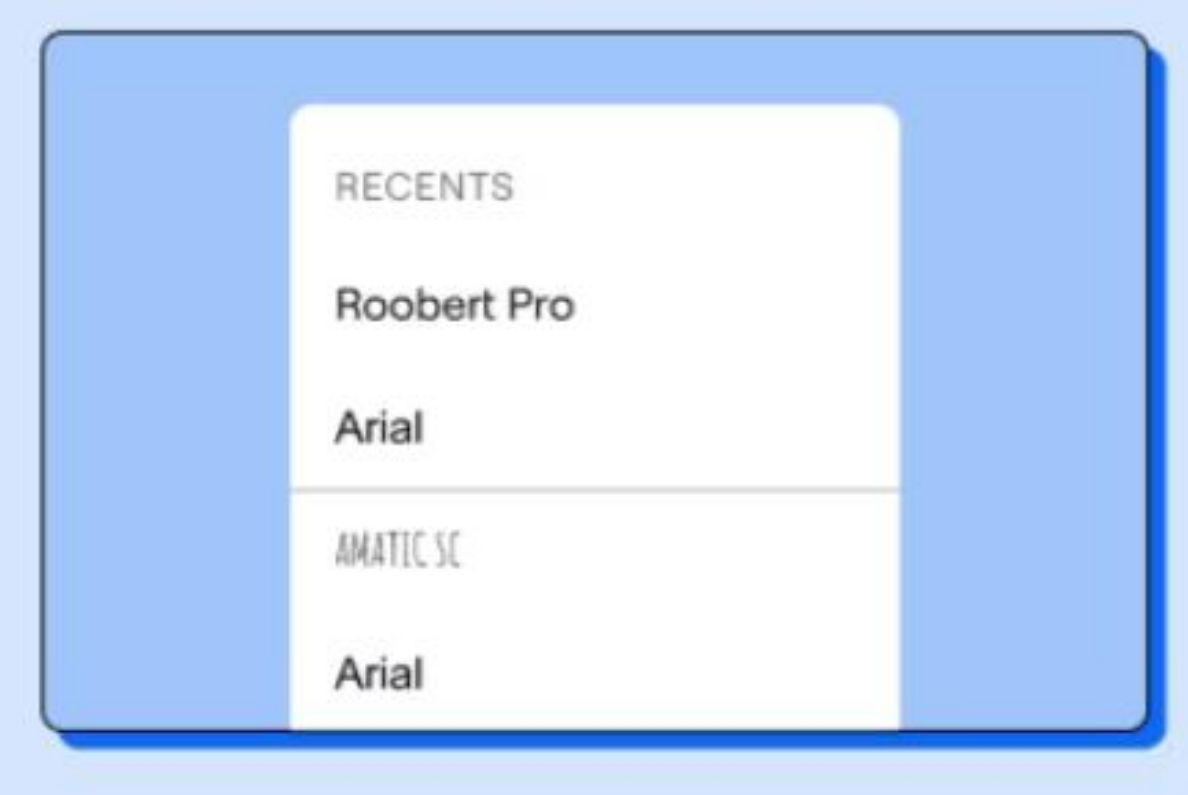


Image Source: Maze Guide on Heuristic Evaluation

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Nielsen's Usability Heuristics with Examples

7. Flexibility and Efficiency of Use

Shortcuts — hidden from novice users — may speed up the interaction for the expert user so that the design can cater to inexperienced and experienced users. Allow users to tailor frequent actions.

Flexible processes can be carried out in different ways so people can pick whichever method works for them.

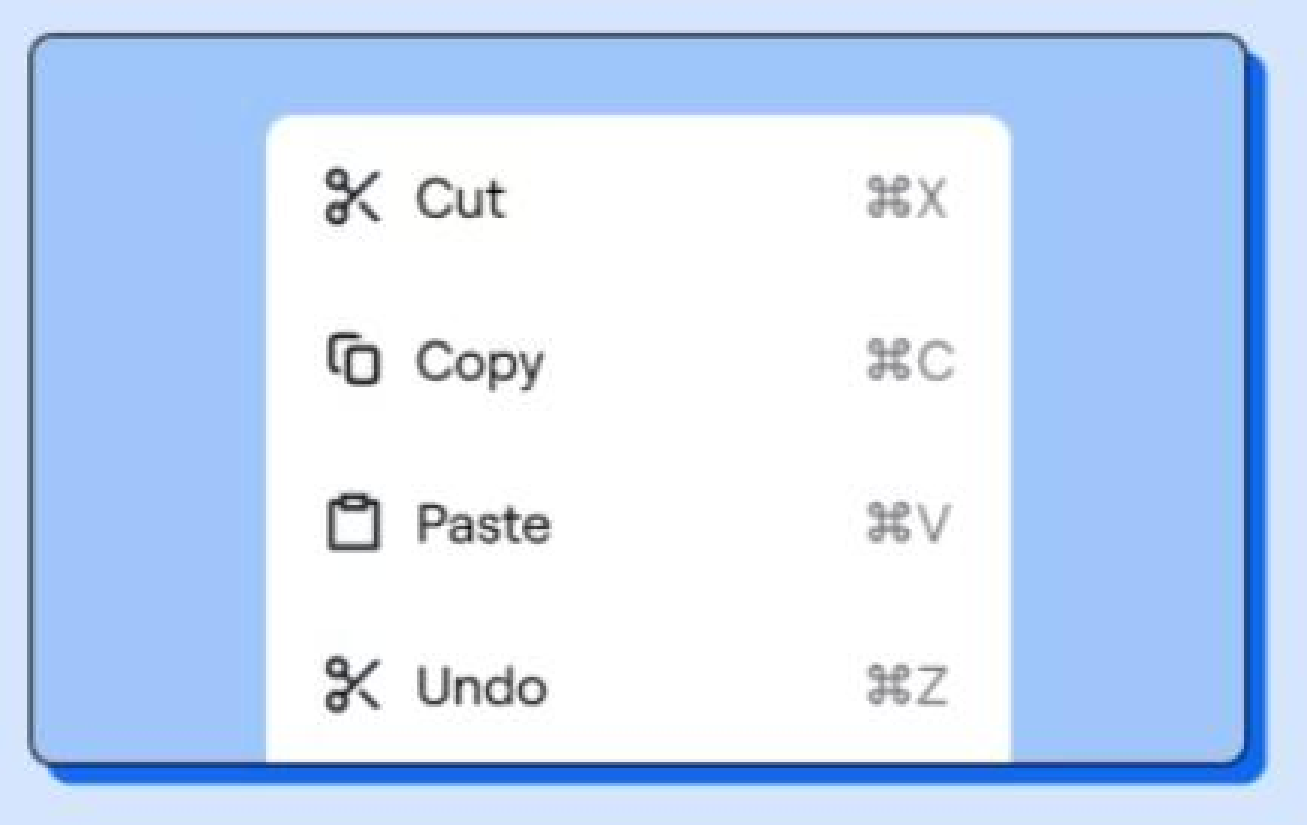


Image Source: Maze Guide on Heuristic Evaluation

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Nielsen's Usability Heuristics with Examples

8. Aesthetic and Minimalist Design

Interfaces should not contain information that is irrelevant or rarely needed. Every extra unit of information in an interface competes with the relevant information units and diminishes their relative visibility.

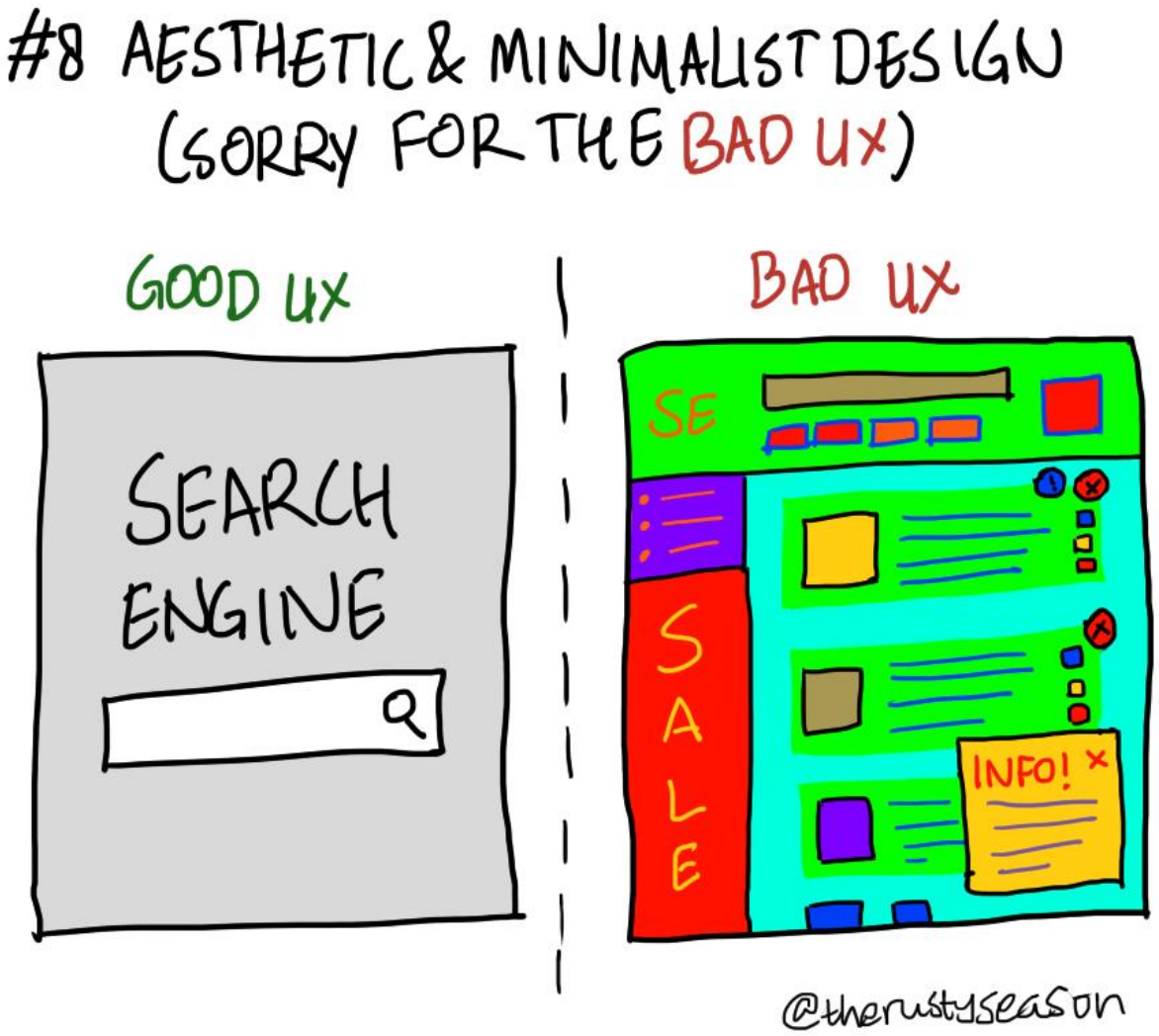


Image Source: [Pratama Naufal, "Jakob Nielsen's 10 Usability Heuristics: Experiences in Comic," Medium]

Nielsen's Usability Heuristics with Examples

9. Help Users Recognize, Diagnose, and Recover from Errors

Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

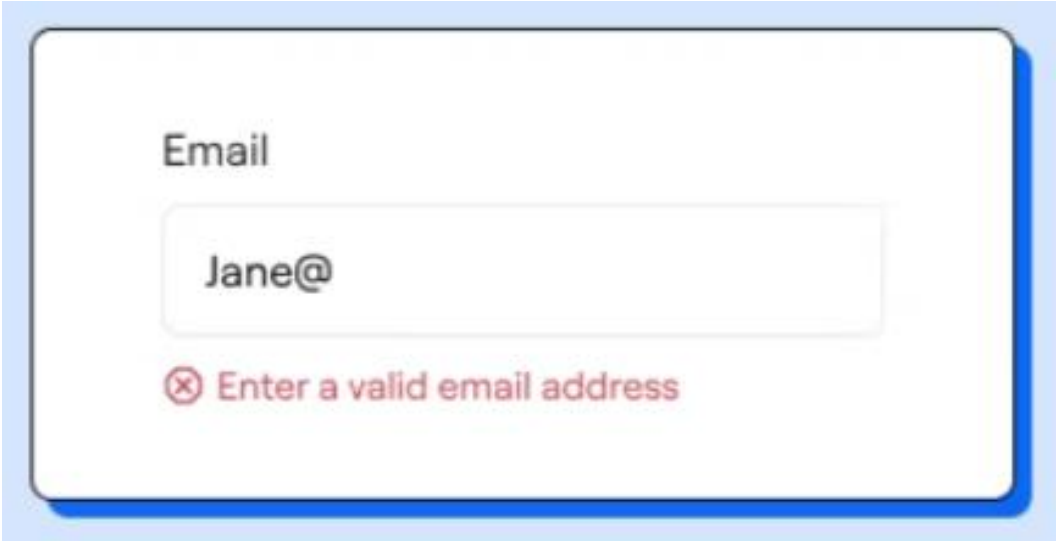


Image Source: Maze Guide on Heuristic Evaluation

Nielsen's Usability Heuristics with Examples

10. Help and Documentation

Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.

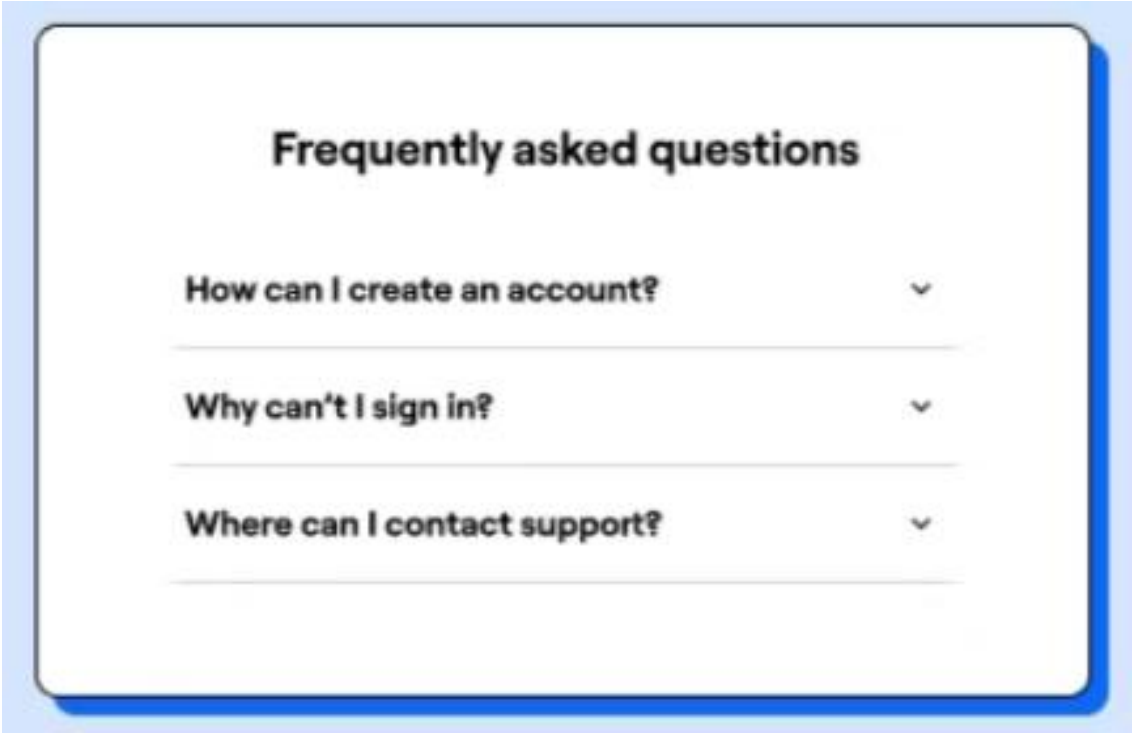
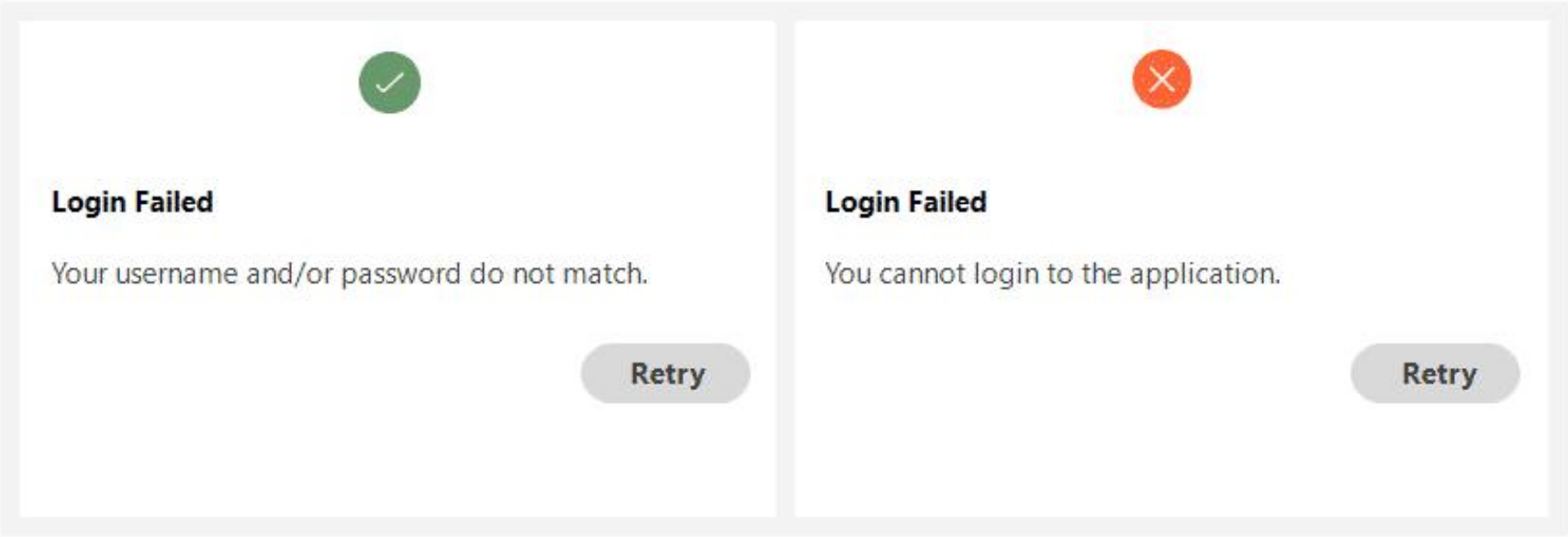


Image Source: Maze Guide on Heuristic Evaluation

Shneiderman's Eight Golden Rules with Examples

Login Failure Example



The reason for login failure is clear and user can enter correct username and password.

The reason of login failure is not clear. User cannot find a solution without hit and trial.

Image Source:[Shneiderman's Eight Golden Rules of Interface Design," UserPeek]

Additional Reading / Resource:

1. "Shneiderman's Eight Golden Rules of Interface Design," UserPeek
2. "Applying the 8 Golden Rules of User Interface Design," UX Matters

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Tips for Using Color Palettes in Web Prototypes

A website color scheme is **a collection of colors that are used together on a website**.
When choosing a color scheme, it is important to think about the mood you want your website to convey.

- **Visual Impact**: Colors influence the mood and tone of your website, shaping how visitors perceive your project
- **Brand Consistency**: A well-chosen color palette ensures your website aligns with your project identity and goals
- **User Experience**: The right use of color enhances readability, guides user actions, and improves navigation.

Good Examples	Bad Examples
Black on green is a good example	Yellow on green is a bad example
Black on yellow is a good example	White on yellow is a bad example
White on orange is a good example	Purple on black is a bad example
Yellow on black is a good example	Blue on gray is a bad example

Image Source:[“The Ultimate Guide to Color Contrast for Websites,” TheGuideX]

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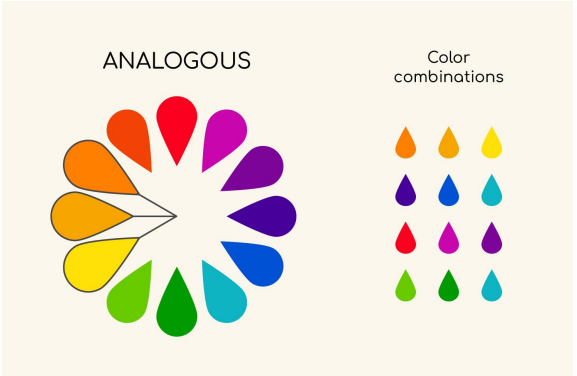
Web Prototype Additional Tips – Color Palettes

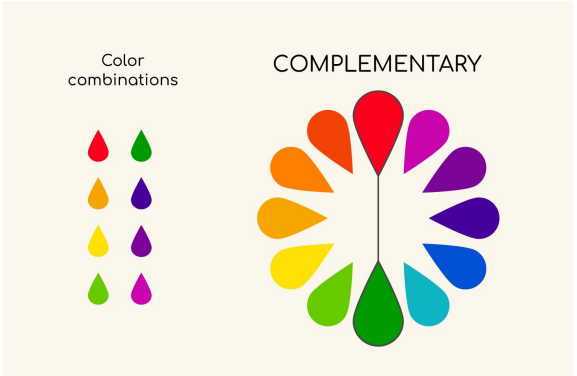
Steps to Creating a Color Palette

1. **Start with Your Project's Theme Colors**:
   - Choose colors that reflect your project's identity and purpose.
   - *Example*: Use your project's logo or central theme colors as a base.

2. **Understand Color Theory**:
   - **Primary, Secondary, and Tertiary Colors**: Know how these relate.
   - **Color Harmony**: Use tools like analogous, complementary, and triadic schemes tocreate a balanced palette.
     - Analogous Color Scheme: This scheme uses colors that are next to each other on the color wheel (Comfortable, lack of contrast)
     - Complementary Color Scheme: This scheme uses colors that are opposite each other on the color wheel (High contrast, need balance)
     - Triadic Color Scheme: This scheme uses three colors that are evenly spaced around the color wheel (Strong contrast)

3. **Consider Color Psychology**:
   - Select colors that evoke the desired emotions for your project's goals (e.g., blue for trust, red for excitement).





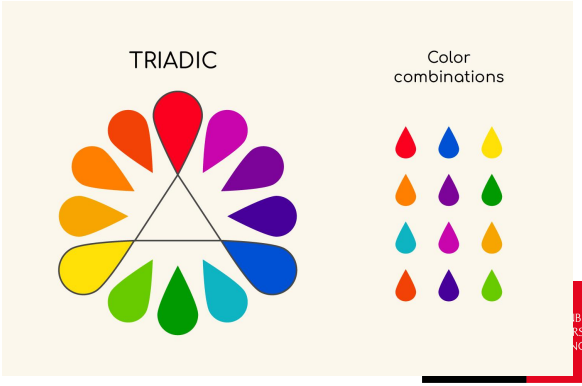


Image Source ["Rules of Color Combination," Amadine,]

Web Prototype Additional Tips – Color Palettes

Consider Color Psychology



Image Source [How to Choose the Right Website Color Scheme," WordStream]

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Web Prototype Additional Tips – Color Palettes

Consider Color Psychology



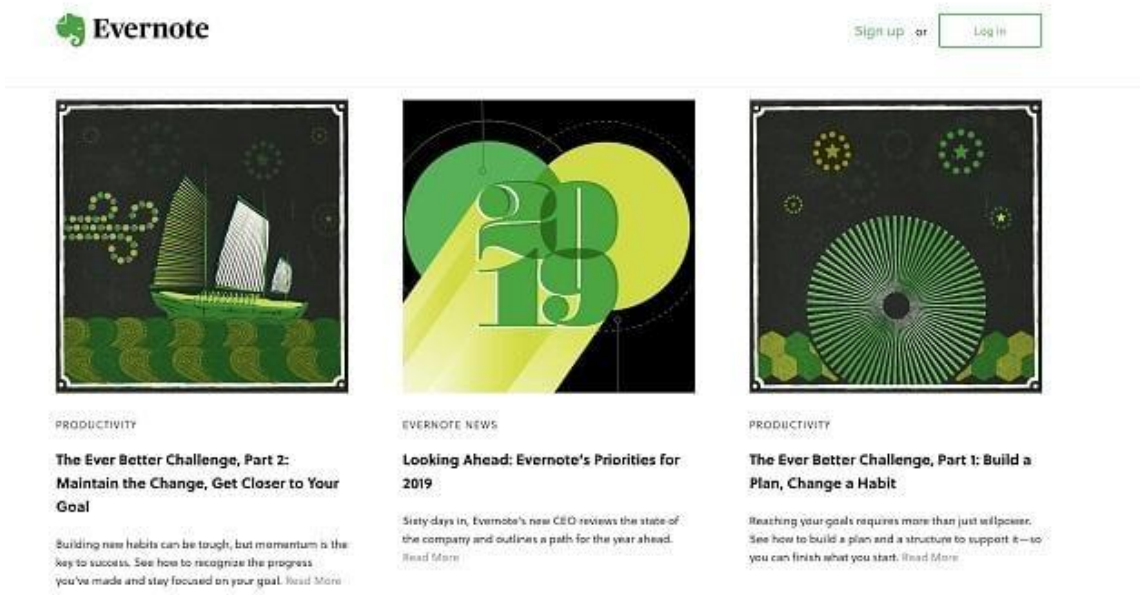


Image Source [How to Choose the Right Website Color Scheme," WordStream]

Web Prototype Additional Tips – Color Palettes

Decide on Primary and Secondary Colors

Primary Colors: The main colors that define the look and feel of your project, aligned with its core identity.
Secondary Colors: Used sparingly to highlight important elements like buttons or links.
Neutral Colors: Provide a balanced background, complementing the primary and secondary colors, typically used for text and backgrounds.

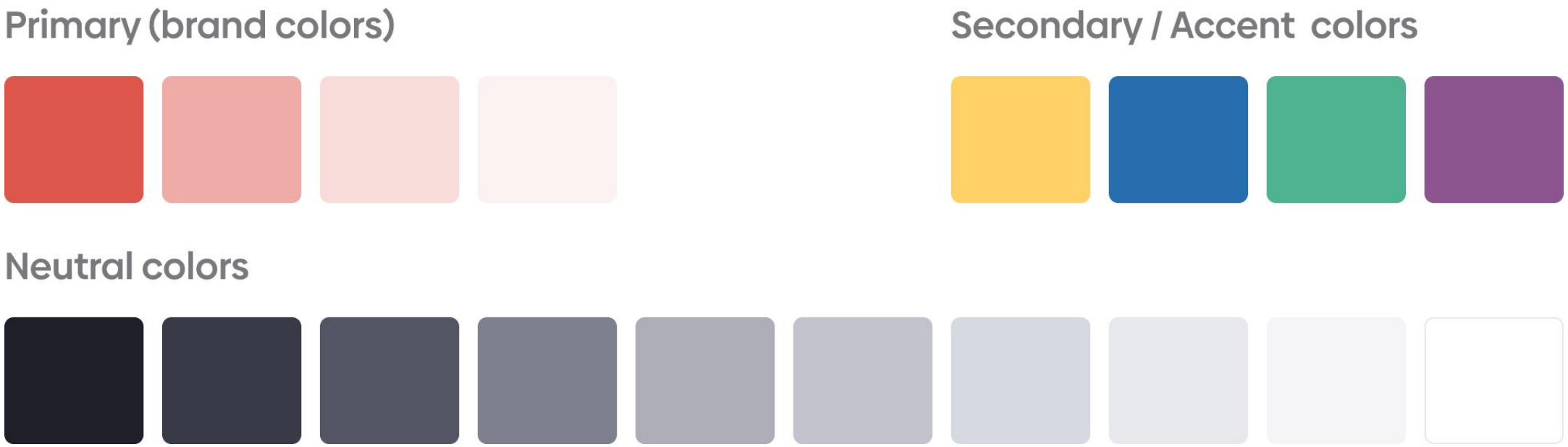


Image Source [“A Guide to Colors in Design Systems,” Supercharge]

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Thanks!

BUT as a developer, picking colors and dealing with design? NOT my thing

I'll stick to code, that's it!



Image Source ["Thank You Emoji," 1000 Logos]

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Image Source [“What to Do With All This Free Time,” Make a Meme]



Image Source [“Sad Will Smith Meme,” Memix]

Web Prototype Additional Tips – Useful Tools

Color Tools:

- Coolors: Quick palette generation.
- Color Hunt: Browse popular palettes.
- My Color Space: Create harmonious color palettes based on a base color.

Tips:

- Limit Your Palette: Stick to 3-5 main colors.
- Use Neutral Colors: Balance bold colors with whites, grays, or blacks.

Additional Resource to Read:

- Complete Guide To Website Color Schemes: Psychology & Examples

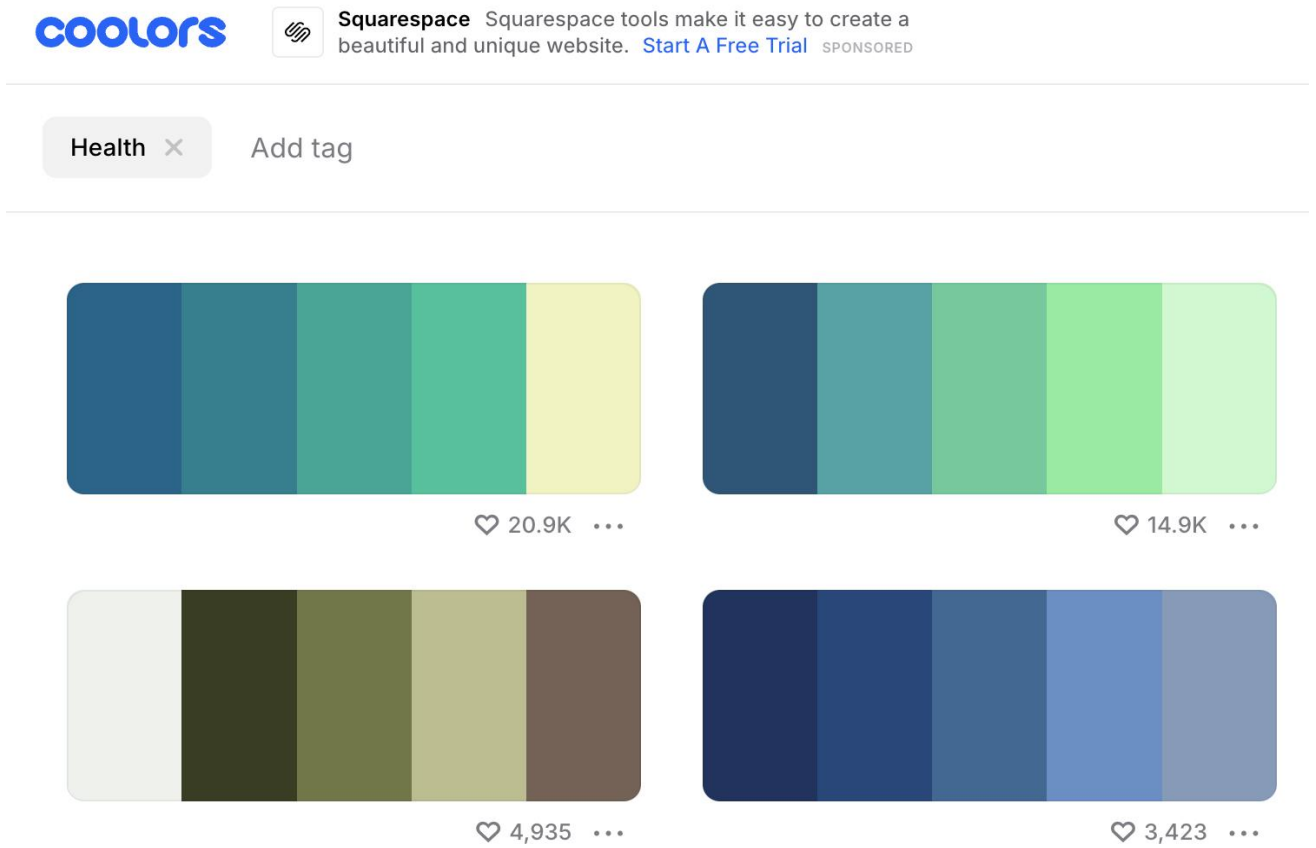


Image Source [Coolors - The Super Fast Color Schemes Generator]

Any Questions?



Image Source:[Top 50 UX Design Memes on the Internet," Uxeria]